

Parle Tilak Vidyalaya Association's
MULUND COLLEGE OF COMMERCE (Autonomous)

F.Y. BBA Semester I

Examination of November 2024

Subject: Practical Approach to Mathematics

Marks: 30

Date: 28/11/2024

Time: 1 Hour

Q1. Attempt either (a) and (b) OR (c) and (d) of the following.

a) Find the sum if I deposit Rs. 25000 at 7.8% for 8 years (08)

b) Find Future value of annuity of Rs. 50000 payable yearly for 5 years at 8%. Per annum. (07)

OR

c) I bought a house worth Rs. 3000000 on loan payable in 4 years in monthly instalments. The rate of interest was 8.4% per annum. What will be the EMI? (08)

d) Find Extra amount receivable Rs.10000 at 8%.pa compounded half yearly for 4 years (07)

Q2 Attempt either (a) and (b) OR (c) and (d) of the following

a) If $A = \begin{bmatrix} 1 & 4 & 2 \\ 3 & 1 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 4 \\ 1 & 5 \\ 1 & 1 \end{bmatrix}$ find $A' + B \wedge B' + A$ (08)

b) if $A = \begin{bmatrix} -2 & 2 & -2 \\ 3 & -3 & 3 \\ 5 & -5 & 5 \end{bmatrix}$ show that $A^2 = A$ (07)

OR

c) Find A^{-1} if $A = \begin{bmatrix} 5 & 1 & 0 \\ 4 & -1 & 3 \\ 3 & 3 & 3 \end{bmatrix}$ (08)

d) i) Find the value of $\begin{vmatrix} 51 & 52 & 53 \\ 54 & 54 & 56 \\ 56 & 57 & 58 \end{vmatrix}$ (04)

ii) Find the value of $\begin{vmatrix} 1 & 4 & 9 \\ 2 & 6 & 15 \\ 3 & 12 & 27 \end{vmatrix}$ without expanding. (03)